Hajar Elsayed Elbehairy

+201003554652 hagarelbehairy3@gmail.com Cairo, Egypt
Hajar Elbehairy Hajar Elbehairy Hajar Elbehairy

PROFILE

Passionate and eager AI & Data Science Engineer with a strong foundation in machine learning, natural language processing, computer vision, and automation, seeking to apply acquired skills, expand expertise, and explore practical horizons in the field of AI and Data Science.

EDUCATION

Postgraduate Diploma of AI-Pro, 9-Month Program **Oct. 2024 – July. 2025**

Information Technology Institute (ITI), Egypt

The program core focuses on ML pipeline, starting from data ETL, all the way to model selection, hyper-parameter tuning, and validation. Moreover, the program includes additional courses related to big data, AWS services, etc.

B.Sc. in Artificial Intelligence **Sep. 2020 – Jul. 2024**

kafrElshiekh University, Egypt

- GPA : 3.75

INTERNSHIPS

Generative AI Intern | Digital Egypt Pioneers Initiative (DEPI) ***July 2024 – Oct 2024***

- Built and deployed AI and Generative AI models using MLOps tools like MLflow and Hugging Face, including developing GANs for content generation and NLP models with attention mechanisms.
- Designed and optimized prompt engineering strategies for large language models (LLMs) to enhance text generation and processing

Artificial Intelligence Training | Samsung Innovation Campus **August 2023 – Oct2023**

- Studied AI fundamentals, machine learning concepts, probability, statistics, and essential mathematics for data science with hands-on Python exercises.
- Applied data processing techniques using NumPy and Pandas, and developed supervised machine learning models for both structured and unstructured data.

PUBLICATIONS

"Human Gait Recognition for Security Systems" - International Technology and Artificial Intelligence Forum (ITAF), Canadian International College (Oct 2024)

- Research presented at ITAF 2024 International Conference "Foresights of Technology and Management within Emerging Developments"

RELEVANT PROJECTS

Video Summarizer (ITI Graduation Project) **Project Repo**

Project aims to build a retrieval-based question-answering system that leverages Large Language Models (LLMs) to generate answers grounded in custom document data.

• **Models**: CLIP, RAG, Whisper (STT), Tesseract OCR, Image Captioning
• **Tools**: Python, Whisper API, FFmpeg, Pydub, OpenCV

Smart HR Recuiter **Project Repo**

Project aims to build a retrieval-based question-answering system that leverages Large Language Models (LLMs) to generate answers grounded in custom document data.

• **Models**: FAISS – Hugging Face Embeddings – RetrievalQA – LangChain LLMs
• **Tools**: Python – LangChain – FAISS – Hugging Face – PyPDF2 – Streamlit

Tasks Scheduling Manager (Agentic AI Project) **Project Repo**

Project aims to automate team task coordination using agent-based architecture with CrewAI, enabling dynamic role-based delegation and execution of work plans using LLMs.

• **Models**: CrewAI Agents – Task & Crew orchestration -calling LLMs
• **Tools**: Python – CrewAI – LangChain – Geminie API – Streamlit – Pydantic

Music and Poem Generation Based on Sentiment Analysis **Project Repo**

Project aims to generate music and poetry that reflect the sentiment of input text using sentiment analysis and generative models.

• **Models**: TextBlob – Huggingface – Music21 – GPT-style poem generation
• **Tools**: Python – TextBlob – Music21 – Streamlit – NLTK – Matplotlib

Deep Learning for Computer Vision — Project Repo

Project aims to apply deep learning techniques to various computer vision tasks, including image classification and semantic segmentation, using custom CNNs, transfer learning, and U-Net architecture.
• **Models**: CNN – Transfer Learning (e.g., VGG, ResNet) – U-Net for Segmentation – Data Augmentation
• **Tools**: Python – TensorFlow – Keras – NumPy – Pandas – Matplotlib

Credit Card Approval Prediction — Project Repo

Project aims to predict whether an applicant will be approved for a credit card using machine learning classification algorithms on structured financial data.
• **Models**: Logistic Regression – Decision Tree – Random Forest
• **Tools**: Python – Scikit-learn – Pandas – NumPy – Matplotlib – Seaborn

Text Clustering Project — Project Repo

Project aims to discover natural groupings in large-scale text data using advanced NLP techniques, topic modeling, and clustering algorithms.
• **Models**: TF-IDF – Latent Dirichlet Allocation (LDA) – K-means – Agglomerative Clustering – PCA – UMAP
• **Tools**: Python – Scikit-learn – Pandas – NLTK – Matplotlib – Seaborn – UMAP – LDA

Graduation Project

A dual-system AI project focused on enhancing security through biometric gait analysis and real-time vehicle identification.

Gait Recognition Security System

Implemented DeepLabV3 for silhouette extraction and Gait Energy Image (GEI) analysis to identify individuals based on their walking patterns.

Arabic License Plate Recognition System(My Role)

Built a real-time detection and recognition system using YOLOv8 for vehicle localization and a custom character detection pipeline for recognizing Arabic license plates.

TECHNICAL SKILLS

- Programming & Problem Solving: Python, C, C++, SQL, NoSQL, Problem Solving
- Mathematics & Statistics: Linear Algebra, Probability, Statistics, Numerical Optimization
- Data Analysis & Visualization: Exploratory Data Analysis (EDA), Data Processing, Pandas, NumPy, Matplotlib, Seaborn, Plotly, Dash, Power BI, Data Cleaning, Preprocessing
- Machine Learning & AI: Supervised & Unsupervised Learning, Deep Learning, NLP, Recommender Systems, Time Series Forecasting, Transformers, Pre-trained Models, MLOps,A/B Testing
- Deep Learning & Frameworks: TensorFlow, Keras, PyTorch, CNN, RNN, LSTM
- Computer Vision: Image Processing, Object Detection, OpenCV, YOLO, MediaPipe,RCNN,Vedio tracking
- GenAI & Agentic AI: LLMs, Prompt Engineering, RAG, LangChain, CrewAI,LangGraph,ollama open source model ,Groq,Agent AI
- Tools & Frameworks: Git, GitHub, Docker, Streamlit, CI/CD,Hugging face,AWS,Spark,PySpark
- Software Engineering & Data Management: OOP, Data Structures & Algorithms, Database Design
- Automation & RPA: UiPath, n8n

SOFT SKILLS

Collaboration, Time Management, Problem-Solving,Team Work, Presentation, Research, Decision Making

EXTRACURRICULAR ACTIVITIES

IEEE Volunteer – Computer Vision Community ***Oct 2022***
IEEE Volunteer – Web Development Community ***Jan 2023***